What Makes Medicinal Leaf Classifiers Transfer from Studio to Field?
A Controlled Study of Leaf Location, Localised Pooling and Measured Geometry

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Abstract

Medicinal-leaf classifiers report near-perfect accuracy on studio photographs of single leaves, but how they transfer to field photographs, and why, is largely unknown. We study this on the 30-species Medicinal Leaf Dataset with a de-duplicated, leaf-grouped protocol, testing on 2,000 DIMPSAR field photographs of 20 shared species. Holding backbone, data and training fixed, we climb a ladder of matched models: a plain classifier, an extra branch on mid-level features, the same branch pooled over the leaf mask, localised pooling over image strips or object-centred sweep slabs, and *sweep-valuation attention* (SVA), a module we introduce whose slab tokens also carry the area and boundary length of learned, appearance-weighted cubical complexes of the blade and venation. Clean accuracy is saturated for every model. On raw field photographs, ConvNeXt-T (5 folds $\times$ up to 5 seeds, corrected resampled tests, Holm correction) gains 12–14 points over a plain model (33.6%). The gain has three parts: mid-level features (+3.5, not significant), knowing where the leaf is (+4.0, which vanishes once the background is whitened) and pooling locally within the leaf (+4.6 to +6.2). Leaf location and localisation persist under strong background-compositing augmentation, which raises every model (best 55%) but replaces neither. Measured geometry adds little if anything: SVA (47.4%) is statistically tied with strip pooling (46.8%) and its own measurement-free variant, and fusing the same measurements globally is equivalent to the plain model within ± 3 points. The gains are small on EfficientNet-B0 and absent on ViTs; the BioCLIP 2 foundation model reaches 81% without any module, whereas a general self-supervised DINOv2 ViT reaches 52%. A pre-registered replication on LeafSnap30, whose field photographs show leaves on plain paper, confirms two of four predictions (no leaf-location effect, no benefit of measured geometry); a smaller total gain holds for strip pooling (+4.1 points) but not for SVA, and the localisation gain (+2.5) is not significant. Extending it to five seeds after the fact makes both total gains significant (strip +5.7, SVA +4.0), while localisation (+3.3) still is not. Code, all run outputs and the probes are released at `https://github.com/Devguru-codes/research_paper_leaf`.

1 Introduction

Medicinal plants are identified in the field mostly from their leaves, and errors are not equally costly: confusing two culinary herbs is harmless, but confusing a remedy with a toxic look-alike, such as *Nerium oleander* with other Apocynaceae, can be dangerous [1]; our field test set contains no such pair, so we cannot evaluate this risk directly (section 7). Deep networks report 97.5–99.7% accuracy on Indian medicinal leaf benchmarks [2, 3]. Those numbers, however, are obtained on single leaves photographed on white backgrounds, typically with random image-level splits. They say little about how the models behave when the same species is photographed in the field, against soil, tiles or cloth, with a phone camera, which is where such a classifier would be used.

Botanists describe a leaf by its geometry: how wide the blade is along its length, where it tapers, how ragged its margin is, how its veins branch. These characters are measured *at particular places* on the leaf. Standard CNN and ViT classifiers average their features over the whole image and have no built-in notion of where the leaf is, let alone of localised geometry. A natural hypothesis is therefore that architectures which measure leaf geometry locally, and look at the image where they measure, should transfer better from studio to field. Recent work fuses topological and morphological summaries with deep features [4–7], but usually as global descriptors, and without controls that would tell whether the geometry itself is what helps.

We test this hypothesis with a ladder of matched models that share backbone, data, vein head and training, and differ in one ingredient at a time: (1) a plain classifier; (2) an extra branch on mid-level patch features; (3) the same branch pooled over the leaf mask, i.e. knowing *where* the leaf is; (4) localised pooling over strips of the image or over slabs of an object-centred directional sweep; and (5) *sweep-valuation attention* (SVA), which we introduce as the most structured member of the family: for each slab of the sweep it measures how much area and boundary length a learned, appearance-weighted cubical complex of the blade and venation adds there, and lets that measurement attend to the image patches of the same slab. A global fusion of the same measurements tests whether geometry helps without localisation.

The answer is clear and partly negative. On ConvNeXt-T the studio-to-field gain over a plain model is large (12–14 points) and splits into three parts: mid-level features (small), knowing where the leaf is (which matters only when the background is present), and pooling locally within the leaf. Measured geometry, whether fused globally or attached to localised tokens, adds little if anything. The gain is small on EfficientNet-B0, absent on ViTs, and irrelevant for a biology foundation model, which transfers far better on its own.

Contributions.

1. **A controlled decomposition of studio-to-field transfer** for medicinal leaves on five backbones, with up to 5 folds $\times$ 5 seeds, planned and post-hoc contrast families with Holm correction, equivalence tests and a bootstrap over runs and field species. Leaf location (+4.0) and localised pooling (+4.6 to +6.2) are the two significant ingredients on raw field photographs of ImageNet ConvNeXt-T, and both add to strong background-compositing augmentation. A pre-registered replication on LeafSnap30 tests the account on a second studio-to-field pair.

2. **Sweep-valuation attention**, a differentiable, backbone-agnostic module that turns a directional filtration of appearance-weighted cubical complexes into slab tokens, and a careful negative result: its measured geometry does not detectably improve transfer beyond localised pooling, its slab-restricted attention can be replaced by global attention, and Euler-characteristic tokens make models sensitive to topology but transfer worse.

3. **A leakage-audited protocol and probes** for the Medicinal Leaf Dataset: duplicate and reshot grouping, a corruption suite (MedLeaf-C), a leaf cue-conflict set, a perceptually matched topology-destruction test and a checked mapping of 20 DIMPSAR field species.

Code, splits, probes, statistics and the outputs of all training runs are released.

2 Related work

Medicinal leaf identification. Work on Indian medicinal leaves has moved from hand-crafted shape, colour and texture descriptors to transfer learning. DeepHerb combined Xception features

with SVMs [2]; later CNN, hybrid CNN–transformer and feature-fusion models report 98.9–99.7% on this and related datasets [3], typically with random image-level splits. Closest to our biological motivation, a dual-branch network fuses an RGB branch with a branch on skeletonised veins by late concatenation [8]. A cascaded taxonomic classifier targets open-world medicinal plants [9]. We reproduce DeepHerb-style and dual-branch pipelines under our leakage-audited protocol.

Studio-to-field shift in plant recognition. The gap between curated and field images is well documented for plant disease recognition: a model trained on the laboratory images of PlantVillage reached 99.4% on held-out laboratory images but about 31% on images taken under other conditions [10], and the background alone of those images carries enough signal to classify them [11]. For medicinal leaves, field evaluation is rare. We measure this gap for a medicinal leaf benchmark under a controlled protocol and ask which architectural ingredients close it.

Euler characteristic transforms. The ECT and the persistent homology transform encode a shape through sublevel-set topology along all directions [12]. DECT makes the ECT differentiable by sigmoid relaxation and learns directions end-to-end [13]. WECT weights cells by image intensity [14]. Blaser et al. encode the ECT as per-direction token sequences, including convolutional and complex-valued variants that are equivariant to planar rotations [7]. For plants, Ayub et al. show that ECT images of *blade contours* classify complex leaf shapes and even predict venation from blade geometry [15]. Their ECT is computed on the outline only; venation is a prediction target, and appearance is not used. We build on DECT's relaxation and on these encodings. We do not claim weighted, tokenised or rotation-equivariant ECTs, or ECTs of leaves, as contributions.

Topology meets attention. Topological summaries are increasingly fused with deep features. PHG-Net injects persistence-diagram features into CNN/transformer classifiers [4]; TDA-ViT fuses topological and ViT features [5]; GraPHFormer aligns graph topology and vision representations [16]. Gu et al. process multi-filtration cubical persistence diagrams with cross-attention and fuse the embedding with CNN/transformer features [6]. TopoFormer slices graphs along filtrations into attention-friendly topological token sequences [17]. For time series, Faghihi and Saki add persistent-homology and directional Euler statistics of each input window as additive biases on attention logits [18]; the topology summarises the whole window rather than defining where each query may attend. In these methods the topological signal enters as a *global* descriptor or as tokens among themselves, so where the visual model looks is not tied to where the topology was measured. SVA differs in exactly this respect: each filtration slab yields both a valuation increment and the spatial support of that token's attention. Our controls (global fusion of the same valuations; slab attention without valuations; mask and strip pooling) separate these ingredients, and we find that the localisation, not the measured topology or geometry, carries the effect.

Intrinsic volumes and directional pooling. Area, boundary length and Euler characteristic (the planar intrinsic volumes, or Minkowski functionals) are classical morphological descriptors with efficient digital estimators [19, 20], and the perimeter of a planar shape can be recovered from its ECT [21]; by Hadwiger's theorem they span all continuous motion-invariant valuations [22, 23]. We use them as filtration increments along a sweep, not as global descriptors, and do not claim them as new features. Strip pooling [24] and axial attention aggregate features along image rows and columns. Our strip control instantiates this family; unlike SVA, its strips are fixed to the image axes, carry no measured geometry and are not defined by a filtration of the object.

Shape versus texture. ImageNet CNNs are biased towards texture [25]. We adapt the cue-conflict paradigm to leaves (silhouette of one species filled with the lamina of another) and add a topology-destruction probe, in order to test what the models rely on.

3 Data and leakage-audited protocol

Dataset. We use the segmented subset of the Medicinal Leaf Dataset of Roopashree and Anitha [26] (CC BY 4.0): 1,835 RGB photographs (1600 × 1200) of single leaves from 30 Indian medicinal species on a white background, with 34–122 images per species (3.6× imbalance). The leaf covers 4.4–68.5% of the frame (median 34.8%). The species span 28 genera, 19 families and 12 orders of the APG IV system, with genuine sibling groups (e.g. Apocynaceae: *Nerium*, *Tabernaemontana*, *Carissa*; Myrtaceae: *Psidium* and two *Syzygium*; Moraceae: two *Ficus* and *Artocarpus*; Lamiaceae: *Mentha*, *Ocimum*, *Plectranthus*); see table 5.

Audit. Reported accuracies on this dataset are near saturation (97.5–99.7% [2, 3]) and use random image-level splits. Our audit found three issues. (i) *Exact duplicates*: 10 pairs of identical photographs saved under different names, all in *Carissa carandas* (1,825 images remain after removal). (ii) *Re-shots of the same physical leaf*: some leaves were photographed more than once, at times under different illumination or orientation. Hashing alone cannot separate these from merely similar leaves (e.g. lanceolate *Alpinia* silhouettes), so we link two images when their raw-frame 1024-bit dHash distance is ≤ 20, or when SIFT matching with RANSAC yields ≥ 40 geometrically consistent inliers whose similarity transform has scale in $[0.8, 1.25]$ and whose convex hull covers $\geq 25\%$ of the leaf. These criteria were set by visual inspection and reject hub matches caused by repetitive texture (e.g. *Mentha*). The resulting 22 multi-image groups (45 images, at most 3 per group) are kept intact across splits. Over-grouping is conservative: it constrains splits but cannot leak. (iii) *Filename collisions*: *Psidium guajava* and *Punica granatum* share the `PG-S-xxx` prefix (65 colliding names), so files must be indexed by class and name.

Protocol. All results use stratified, group-aware 5-fold cross-validation on the 1,825 de-duplicated images. Inside each training part, one group-aware stratified holdout ($\approx 1/6$) is used for monitoring only. Epoch count and hyper-parameters are fixed in advance, with no model selection on test or calibration data. With random image-level splits, 1.4–3.0% of test images per fold have a duplicate or re-shot in the training part (section 6.8).

Preprocessing. The leaf mask is obtained by thresholding the white background (all channels < 235) followed by morphological cleaning. Each image is cropped to a square centred on the leaf whose inscribed circle contains the whole leaf, so that any rotation about the centre is lossless, and resized to 448^2 (training input 384^2). Vein pseudo-labels are multiscale Sato ridge responses on CLAHE-normalised luminance inside the eroded mask.

Cross-domain test set. For transfer to field photographs we use DIMPSAR [27] (CC BY 4.0), whose single-leaf subset was captured with smartphones against varied backgrounds (tiles, soil, cloth, paper). Vernacular folder names were mapped to species and checked visually. Twenty species overlap ours; note that Kannada *Arali* denotes *Ficus religiosa* (peepal), not *Nerium oleander*. Leaves are localised with our own masker, a ConvNeXt-T FPN trained only on our leaves composited onto DTD textures [28] and procedural backgrounds; no third-party segmentation model is used. The masker defines the field test set for every model: each photograph is cropped to a square

around the largest predicted leaf component, exactly as the studio images are cropped around their leaf, and photographs in which it finds no plausible leaf (predicted leaf share outside 0.5–95%) are excluded. This removes 118 of the 2,118 photographs (5.6%), 49 of them from the compound-leaved *Moringa* and *Azadirachta* (table 7), leaving 2,000. The *raw* condition keeps the background inside the crop; the *masked* condition additionally whitens it with the same mask.

4 Models

4.1 A ladder of matched models

To isolate what matters, we compare seven models that share the backbone, a light vein head on early backbone features with its auxiliary loss, the training recipe and, where applicable, the token width (192) and transformer depth (2). Each step adds one ingredient; all but the first concatenate their output with the backbone's global token before the linear classifier.

1. **plain**: backbone global token only, i.e. top-stage features averaged over the whole image, background included;
2. **patch pooling**: adds a branch on the mid-level (stage-3) patch tokens, an MLP per token averaged over the whole image; the extra features and capacity that all following models share, without any mask;
3. **mask pooling**: the same branch averaged over the leaf mask only; the model knows *where* the leaf is but pools it as a whole;
4. **strip pooling** [24]: mask-weighted means of the patch tokens over every row and column of the patch grid, with learned positions and the transformer; localised pooling along the image axes;
5. **slab-only**: SVA (section 4.2) without measurements, so each token only pools the patches of its slab of an object-centred directional sweep;
6. **SVA**: slab tokens that also carry measured leaf geometry;
7. **global fusion**: the full grid of SVA's measurements flattened through an MLP and fused globally (in the spirit of [4, 6]); the same measurements without localised pooling.

The contrasts are: 2 vs. 1 (*mid-level features*); 3 vs. 2 (*leaf location*); 4, 5 and 6 vs. 3 (*localisation*); 6 vs. 5 and 6 vs. 4 (*measured geometry*, and object-centred slabs vs. image-axis strips); and 7 vs. 1 (*geometry without localisation*). Every model receives the same field crops, centred by our masker (section 3); models 3–7 additionally use the predicted mask inside the network, whereas models 1 and 2 use only the crop. The masker is a second network that costs about as much as the classifier (section E).

4.2 Sweep-valuation attention

SVA is our attempt to give a network explicit, local leaf geometry (fig. 1). Two structural fields are derived from the image: the *blade* (the leaf mask m) and a *learned venation field* v, the vein probability predicted by the vein head, which is pseudo-supervised by a multiscale Sato ridge response [29] and is not an observed vascular structure. On a 96×96 grid in the blade's centroid/RMS

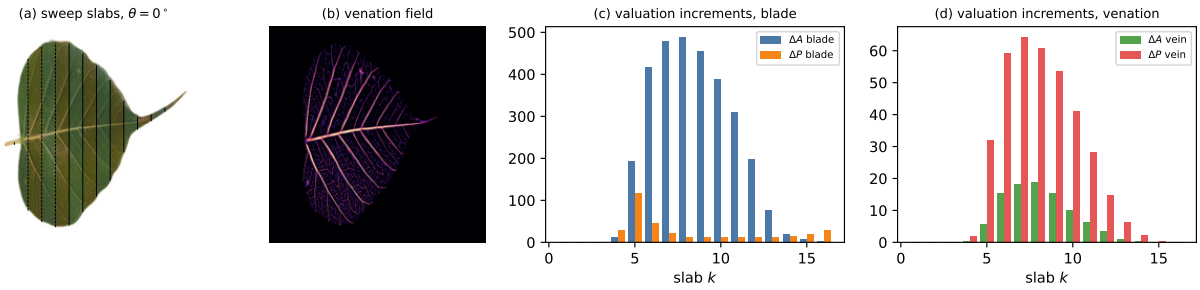


Figure 1: The sweep-valuation signal for one leaf (*Ficus religiosa*) and one of the K=32 directions. (a) The leaf is partitioned into T=16 slabs along θ in its centroid/RMS frame. (b) Ridge-filter vein pseudo-label that supervises the learned venation field. (c,d) Area and perimeter increments per slab for the blade and the venation field, shown here with unit weights; in SVA the cell weights are appearance-conditioned. Each slab's increments form the query of one token, which attends to the image patches of the same slab. The long perimeter in the last blade slabs comes from the drip tip.

frame, each field becomes a weighted cubical complex whose cell weights are conditioned on appearance: per-pixel weights $a_{\mathrm{b}} = m\,\sigma(g_{\mathrm{b}}(f))$ and $a_{\mathrm{v}} = m\,v\,\sigma(g_{\mathrm{v}}(f))$ from mid-level features f and small MLPs g, and each edge and square inherits the minimum weight of its pixels,

$$w_0(p) = a(p), \qquad w_1(e) = \min_{v \in e} a(v), \qquad w_2(q) = \min_{v \in q} a(v), \tag{1}$$

which keeps the weighted complex a fuzzy subcomplex and reduces to the ordinary cubical complex for binary weights. With H_d the weighted number of d-cells, area $A = H_2$, boundary length $P = 2H_1 - 4H_2$ and Euler characteristic $\chi = H_0 - H_1 + H_2$ are the digital intrinsic volumes (section A).

For each of K=32 directions θ, the sweep is cut into T=16 soft slabs with the sigmoid relaxation of DECT [13]. For every slab, SVA records how much area and boundary length of both fields enter it (and how much has entered so far), which describes the local width and boundary complexity of the leaf at that position. Each slab yields one token whose query is built from these measurements and which attends only to the image patches of the same slab; the 512 tokens are mixed by a two-layer transformer and mean-pooled. To the best of our knowledge, no prior work co-registers filtration-derived valuation increments with attention restricted to the corresponding filtration slab (section 2); our experiments show, however, that neither the measurements nor the restriction is what makes the model transfer. The equations, the efficient histogram computation, the Euler and Hadwiger token variants and the rotation property are given in section A.

5 Experimental setup

Backbones and training. We use two ImageNet-1k CNNs (ConvNeXt-T and EfficientNet-B0), ViT-S/16 (AugReg, ImageNet-21k→1k) and the BioCLIP 2 ViT-L/14 visual tower (all but the last two blocks frozen). All models of a backbone share the recipe: AdamW (new modules lr 10^{-3}, weight decay 0.05), 2 warm-up epochs then cosine decay, batch 32, mixed precision, label smoothing 0.1. The backbone lr is 10^{-4} for ConvNeXt-T, 2×10^{-4} for EfficientNet-B0 and 5×10^{-5} for ViT-S and 3×10^{-5} for DINOv2 ViT-S/14 [30], all trained 40 epochs at 384^2; BioCLIP 2 uses lr 3×10^{-5}, 20 epochs and 224^2 input. The recipe was fixed in advance and not tuned per model. Augmentation uses uniform rotation in $[-180°, 180°]$, scale $[0.85, 1.15]$, translation $\pm 5\%$, horizontal flips and mild

colour jitter, applied jointly to image, mask and vein pseudo-label. No early stopping or test-time augmentation is used. Runs use Kaggle T4 GPUs.

Runs. All splits are grouped stratified 5-fold cross-validation over the de-duplicated, leaf-grouped data (section 3). For ConvNeXt-T, plain, patch pooling, mask pooling, strip, slab-only, SVA and global fusion are trained with 5 seeds per fold (25 runs each); the field test is run on all seeds, MedLeaf-C on several seeds but reported on seed 0, the one seed run for every model, and the other probes on seed 0. EfficientNet-B0 uses 3 seeds per fold, ViT-S, BioCLIP 2 and DINOv2 ViT-S/14 one seed per fold. Two further ConvNeXt-T arms test whether the effects depend on augmentation. *Without rotation augmentation*: plain, strip, slab-only and SVA (5 seeds) and mask pooling (3 seeds). *Strong augmentation*: plain, mask pooling, strip and SVA, 3 seeds, with background compositing (with probability 0.8 the white background is replaced by a random procedural texture, blended through the slightly eroded mask), Gaussian blur ($p = 0.3$) and Gaussian noise ($p = 0.3$) on top of the standard augmentation. A *learning-rate check* retrains plain and strip pooling (seed 0, 5 folds) at half and twice the ConvNeXt-T backbone learning rate. The LeafSnap30 replication and its runs are described in section 6.4.

Statistics. The primary outcome is 20-way accuracy on raw DIMPSAR field photographs; the masked photographs are the secondary outcome. Each contrast is the mean difference over matched fold–seed runs, with a 95% interval from the Nadeau–Bengio corrected resampled t-test [31] (test/train ratio 0.3, accounting for overlapping training sets; slightly above the 0.25 of a 5-fold split, hence conservative). The DIMPSAR study was not pre-registered (the LeafSnap30 replication was, section 6.4), and we separate two families of contrasts, each Holm-corrected [32] over both photo types. The *planned* family was fixed before any run: each localised model vs. plain, SVA vs. slab-only and vs. strip pooling (measured geometry), and global fusion vs. plain (12 tests). The *post-hoc* family uses the patch- and mask-pooling controls, which we added after a first analysis to test a suspected confound (10 tests); these contrasts confirm or refute that confound rather than test a prior hypothesis. A difference is called significant only if its Holm-corrected $p < 0.05$. For the geometry and global-fusion contrasts, where the question is whether an ingredient helps at all, we also test equivalence within ± 3 accuracy points with two one-sided corrected t-tests. As a robustness check, a two-way bootstrap that resamples runs and field species (clusters of images) jointly also captures sampling of the fixed field test set. Secondary comparisons (other arms, backbones, probes) are reported with the corrected interval and win counts, without multiplicity claims.

Ablations of SVA. Euler-only and full-Hadwiger tokens; blade-only and vein-only fields; unit cell weights; *permuted* learned weights (the same weight values shuffled within the leaf before computing the valuations); global instead of slab-restricted attention; a coarser sweep (K=16, T=8); and a softer or sharper relaxation ($\kappa \in \{1, 16\}$). All ablations run on seed 0, folds 0–2; the four most informative (unit weights, global attention, coarse sweep, Euler token) are repeated on all five folds and seeds 0–2 and compared with SVA on the same runs.

Probes. All probes are applied to each fold's test images with fixed seeds.

- **Field photographs**: models trained on the studio images are tested on the 2,000 DIMPSAR photographs of the 20 overlapping species (section D), on raw crops with the background kept and on crops whose background was whitened by our masker. Accuracy is 20-way, i.e. restricted to the shared species; this uses the knowledge of which species can occur and is the same for all models.

- **MedLeaf-C**: ten corruptions × three severities: Gaussian noise, blur, JPEG, low resolution, darkening, contrast loss, hue shift (yellowing), desaturation (severity 3 is grayscale), occluders, and pasted procedural/soil-like backgrounds. Masks are the oracle masks transformed consistently.
- **Rotation sweep**: accuracy at $0°, 30°, \ldots, 330°$.
- **Cue conflict**: the silhouette of a test leaf A is filled with the lamina of a test leaf B of another species. B is aligned by principal axes and enlarged until it covers A, so that B's outline contributes no shape cue. Shape bias $= n_A/(n_A + n_B)$ where n_A (n_B) counts predictions of A's (B's) species.
- **Topology destruction**: three interior holes in the blade (exact $\Delta\chi = -3$, verified) versus three boundary bites of equal removed area ($\Delta\chi = 0$, verified), and inpainting on strong veins versus vein-free lamina. Every edit records LPIPS, pixel RMSE and a colour-histogram distance; the response (KL divergence between clean and edited predictions) is compared within LPIPS strata (section F).

6 Results

All numbers below are generated from the run outputs by the released scripts; ± is the standard deviation over runs. Differences are given as mean over matched fold–seed runs, with the number of runs in which the first model wins and 95% intervals as described in section 5.

6.1 In-domain accuracy is saturated

On the de-duplicated, grouped protocol every model on every backbone reaches 99.8–100% clean accuracy and $\geq$ 98.5% accuracy under any rotation (table 1). The studio benchmark therefore cannot separate architectures, and none of the added modules costs clean accuracy. All further results concern robustness and, primarily, transfer to field photographs.

6.2 Decomposing the studio-to-field gain

Figure 2 and table 2 show the main result. A plain ConvNeXt-T trained on studio images reaches 33.6% on raw field photographs; the localised models reach 45.7–47.4%, a gain of 12–14 points (balanced accuracy: 31.4% vs. 42.9–44.6%). The ladder of matched models splits this gain into three parts.

Mid-level features. An extra branch that averages mid-level patch features over the whole image, without any mask, adds +3.5 points on raw photographs (22/25 runs, not significant) and +4.4 on masked ones (25/25, Holm $p < 0.001$).

Leaf location. Averaging the same branch over the leaf mask instead adds a further +4.0 points on raw photographs (25/25, Holm $p = 0.001$) but only +1.3 on masked ones (n.s.). Knowing where the leaf is thus matters when the background is present and hardly at all when it has been whitened, which identifies this part as discounting the field background. On whitened photographs, the gain of mask pooling over plain is almost entirely the mid-level-feature effect.

Localisation. Pooling locally within the leaf adds another 4.6–6.2 points on raw photographs: strip pooling +5.7 (25/25, Holm $p = 0.002$) and slab-only attention +4.6 (24/25, Holm $p = 0.006$) are significant; SVA +6.2 (23/25) is not after correction (Holm $p = 0.11$) because its run-to-run variance is the largest. On masked photographs localisation adds 1.2–1.6 points (n.s.).

Table 1: Main results (%). Clean = in-domain grouped 5-fold accuracy; Rot. min = worst accuracy over 12 rotations; MedLeaf-C = mean over 10 corruptions × 3 severities; Shape bias from the cue-conflict set; Field = 20-way accuracy on DIMPSAR field photographs (auto-masked or raw). ConvNeXt-T: 5 folds × 5 seeds (MedLeaf-C: seed 0, the seed run for every model; Rot. min and shape bias: seed 0); EfficientNet-B0: 5 folds × 3 seeds; ViT-S, DINOv2 and BioCLIP 2: 5 folds, one seed. Patch pooling was run on ConvNeXt-T only, mask pooling on ConvNeXt-T and DINOv2.

Backbone	Model	Clean	Rot. min	MedLeaf-C	Shape bias	Field (mask)	Field (raw)
ConvNeXt-T	Plain	99.9 ± 0.1	99.9 ± 0.2	81.1 ± 1.1	6.5 ± 1.5	35.2 ± 1.1	33.6 ± 1.2
	Global fusion	99.9 ± 0.1	99.7 ± 0.3	81.0 ± 0.8	7.0 ± 1.9	34.6 ± 1.5	33.5 ± 1.1
	Patch pooling	100.0 ± 0.1	–	84.3 ± 0.9	–	39.6 ± 1.1	37.1 ± 2.3
	Mask pooling	100.0 ± 0.1	–	84.7 ± 0.6	–	40.9 ± 1.3	41.1 ± 1.6
	Strip pooling	100.0 ± 0.0	99.9 ± 0.1	87.6 ± 0.8	9.1 ± 1.5	42.4 ± 1.6	46.8 ± 2.2
	Slab-only	100.0 ± 0.1	100.0 ± 0.0	89.4 ± 0.9	7.4 ± 1.9	42.1 ± 1.2	45.7 ± 1.8
	SVA (ours)	100.0 ± 0.1	99.9 ± 0.1	88.6 ± 0.5	9.0 ± 2.3	42.5 ± 2.6	47.4 ± 3.7
EffNet-B0	Plain	99.9 ± 0.1	99.9 ± 0.1	76.9 ± 0.8	–	32.2 ± 1.6	26.8 ± 3.2
	Global fusion	100.0 ± 0.1	99.7 ± 0.2	78.1 ± 1.1	–	31.7 ± 1.1	29.2 ± 2.1
	Strip pooling	99.9 ± 0.1	99.8 ± 0.1	77.2 ± 0.4	–	31.1 ± 1.1	29.1 ± 2.3
	Slab-only	99.9 ± 0.1	99.7 ± 0.2	77.9 ± 0.8	–	30.3 ± 1.4	28.3 ± 2.7
	SVA (ours)	99.9 ± 0.2	99.8 ± 0.2	77.7 ± 1.6	–	31.1 ± 1.4	29.8 ± 2.2
ViT-S/16	Plain	99.9 ± 0.1	99.9 ± 0.1	96.4 ± 0.3	9.7 ± 1.0	51.6 ± 0.4	60.4 ± 0.9
	Global fusion	100.0 ± 0.0	99.9 ± 0.1	96.2 ± 0.6	9.1 ± 1.3	50.1 ± 0.7	58.6 ± 1.8
	Strip pooling	99.9 ± 0.1	99.9 ± 0.1	96.2 ± 0.5	10.3 ± 1.3	51.2 ± 1.1	62.7 ± 1.0
	Slab-only	100.0 ± 0.0	99.8 ± 0.1	96.6 ± 0.5	11.0 ± 1.4	52.0 ± 0.8	64.8 ± 2.1
	SVA (ours)	99.9 ± 0.1	99.9 ± 0.1	96.6 ± 0.3	9.3 ± 1.8	51.9 ± 1.0	60.7 ± 1.4
DINOv2-S/14	Plain	99.8 ± 0.1	–	90.8 ± 0.6	–	45.1 ± 1.6	52.5 ± 2.1
	Mask pooling	100.0 ± 0.0	–	90.9 ± 1.1	–	43.5 ± 2.5	48.0 ± 2.4
	Strip pooling	100.0 ± 0.0	–	91.3 ± 1.1	–	45.9 ± 1.9	53.0 ± 2.1
BioCLIP 2	Plain	99.9 ± 0.1	99.0 ± 0.4	93.8 ± 0.5	14.0 ± 1.3	58.9 ± 0.5	81.4 ± 0.7
	Global fusion	99.9 ± 0.1	99.0 ± 0.4	93.8 ± 0.6	14.3 ± 2.3	59.7 ± 1.1	81.3 ± 0.2
	Strip pooling	99.9 ± 0.1	99.2 ± 0.5	93.8 ± 0.2	14.2 ± 1.7	58.9 ± 0.7	81.5 ± 0.5
	Slab-only	100.0 ± 0.0	98.8 ± 0.5	93.9 ± 0.6	13.9 ± 1.2	60.7 ± 0.7	81.8 ± 1.4
	SVA (ours)	100.0 ± 0.0	98.5 ± 0.5	93.5 ± 0.5	13.7 ± 2.0	59.8 ± 1.1	81.7 ± 0.8

Measured geometry adds little if anything. Fusing the valuations globally, without localisation, is equivalent to plain within ±3 points (-0.1, 13/25; equivalence $p < 10^{-3}$ on raw and $p = 0.006$ on masked photographs). Attached to localised tokens, they give +0.5 over strip pooling (12/25) and +1.7 over slab-only attention (16/25) on raw photographs; neither is significant, and the intervals exclude large benefits but not benefits of up to 3–5 points (equivalence within ±3 is shown only for SVA vs. strip on masked photographs). Object-centred sweep slabs and image-axis strips perform the same. Per species, SVA beats plain on 17 of the 20 species and strip pooling on 8 (table 7). In the planned family, each localised model beats plain by 12–14 points on raw and about 7 on masked photographs (25/25 runs, Holm $p < 10^{-3}$). The two-way bootstrap agrees with the corrected test on every significant contrast. Without the restriction to the 20 shared species (30-way accuracy), every model loses 3–7 points but the ordering is unchanged: plain 30.0%, mask pooling 35.0%, strip pooling 39.5%, SVA 40.5% on raw photographs, and BioCLIP 2 76.2%.

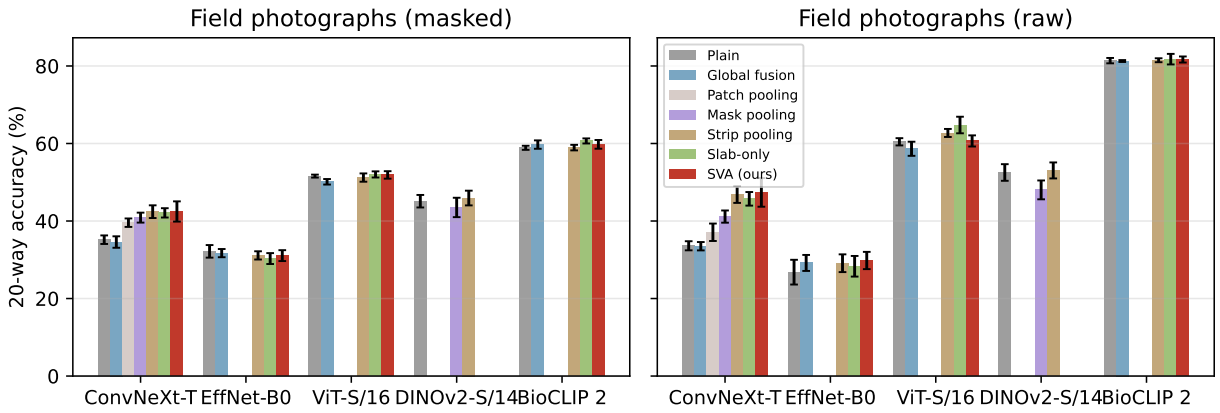


Figure 2: Studio-to-field transfer: 20-way accuracy on DIMPSAR photographs with the background whitened by our masker (left) or kept (right). Error bars: std over runs. Patch pooling was run on ConvNeXt-T only, mask pooling on ConvNeXt-T and DINOv2.

6.3 The gains are not an artefact of augmentation or tuning

Without rotation augmentation (5 folds × 5 seeds), the localised models remain far more rotation-robust than plain (worst-case accuracy 97.5–98.4% vs. 93.2%; fig. 3) and keep their field gain: plain 32.1%, slab-only 43.2%, strip 43.9%, SVA 45.5% on raw photographs (SVA vs. plain +13.5 [+9.7, +17.3], 25/25). SVA is again not distinguishable from strip pooling (+1.6, 16/25) or slab-only attention (+2.4, 20/25). With a single seed these leads had looked much larger (+5.8 and +5.2), which illustrates why we rely on many seeds. Mask pooling, trained on seeds 0–2 of this arm, reproduces the decomposition: it adds +8.2 [+4.5, +11.9] points over plain on raw photographs (15/15), and localised pooling adds a further +3.7 to +7.3 (14–15/15) on raw but no significant gain on masked photographs.

With strong augmentation (background compositing, blur, noise; 3 seeds; table 3), every model improves: plain rises from 33.6% to 42.0% on raw photographs and MedLeaf-C reaches 96–98%. Both architectural parts still add on top. Mask pooling gives +8.0 [+4.7, +11.3] over plain (15/15), so training on composited backgrounds does not replace knowing where the leaf is, and localisation adds a further +3.3 (strip, 15/15) to +5.2 (SVA, 15/15) on raw photographs (none on masked ones). SVA with strong augmentation reaches 55.2%, the best field accuracy of any ImageNet model here, but is again not distinguishable from strip pooling (+1.9, 13/15). The architectural and data-side remedies are therefore complementary.

Across learning rates. A gap this large invites the objection that the plain baseline was under-tuned. We retrained plain and strip pooling (seed 0, 5 folds) at half and twice the backbone learning rate. Plain's raw field accuracy moves only from 34.0% to 35.3% across the three rates (5×10^{-5}, 10^{-4}, 2×10^{-4}), and strip pooling beats it at every rate: +13.5, +12.2 and +8.4 points (5/5 folds each, corrected $p \leq 0.001$); on masked photographs the gap is +4.0 to +6.4. The best plain setting (35.3%) stays below the worst strip setting (43.7%).

6.4 A pre-registered replication on LeafSnap30

To test whether the decomposition carries over to other data, we wrote down four predictions before training any model (public commit `90c47b7`, `paper/preregistration_leafsnap.md`, made on 25 October 2026 about 12 hours before the first LeafSnap run was started) and repeated the study

Table 2: Pre-specified contrasts on field photographs (ConvNeXt-T, 25 matched runs). Δ in accuracy points with runs won; 95% intervals from the corrected resampled t-test over runs and from a bootstrap over runs and field species; p Holm-corrected within each family. †equivalent to zero within ±3 points (two one-sided tests, $p < 0.05$).

Set	Photos	Contrast	Δ (runs won)	95% CI, runs	95% CI, runs+species	Holm p
planned	raw	strip − plain	+13.2 (25/25)	[+10.0, +16.5]	[+7.9, +18.8]	$< 10^{-4}$
		slab-only − plain	+12.1 (25/25)	[+9.4, +14.8]	[+5.4, +18.8]	$< 10^{-4}$
		SVA − plain	+13.8 (25/25)	[+9.3, +18.3]	[+7.3, +20.4]	$< 10^{-4}$
		SVA − slab-only	+1.7 (16/25)	[-3.5, +6.8]	[-2.8, +6.5]	1.00
		SVA − strip	+0.5 (12/25)	[-3.6, +4.7]	[-3.0, +4.5]	1.00
		global fusion − plain	-0.1† (13/25)	[-1.5, +1.3]	[-2.3, +2.4]	1.00
	masked	strip − plain	+7.2 (25/25)	[+4.7, +9.8]	[+3.3, +11.5]	$< 10^{-4}$
		slab-only − plain	+6.9 (25/25)	[+4.8, +9.0]	[+2.3, +11.8]	$< 10^{-4}$
		SVA − plain	+7.3 (25/25)	[+4.2, +10.4]	[+3.8, +11.0]	0.000
		SVA − slab-only	+0.4 (15/25)	[-3.2, +3.9]	[-3.4, +4.4]	1.00
		SVA − strip	+0.1† (15/25)	[-3.4, +3.5]	[-2.9, +3.4]	1.00
		global fusion − plain	-0.6† (8/25)	[-2.4, +1.2]	[-3.4, +2.3]	1.00
post hoc	raw	patch pooling − plain	+3.5 (22/25)	[+0.4, +6.6]	[-0.3, +7.2]	0.15
		mask pooling − patch pooling	+4.0 (25/25)	[+2.2, +5.9]	[+2.2, +6.1]	0.001
		strip − mask pooling	+5.7 (25/25)	[+3.0, +8.4]	[+2.3, +9.2]	0.002
		slab-only − mask pooling	+4.6 (24/25)	[+2.1, +7.0]	[+0.7, +8.9]	0.005
		SVA − mask pooling	+6.2 (23/25)	[+1.2, +11.3]	[+2.4, +10.2]	0.11
	masked	patch pooling − plain	+4.4 (25/25)	[+2.6, +6.2]	[+1.6, +7.4]	0.000
		mask pooling − patch pooling	+1.3 (22/25)	[-0.0, +2.6]	[+0.3, +2.4]	0.22
		strip − mask pooling	+1.5 (21/25)	[-0.7, +3.7]	[-1.4, +4.3]	0.49
		slab-only − mask pooling	+1.2 (20/25)	[-0.7, +3.2]	[-2.4, +5.0]	0.49
		SVA − mask pooling	+1.6 (17/25)	[-1.7, +4.9]	[-1.7, +4.9]	0.49

on LeafSnap30 [33, 34]: 30 North-American tree species, 5,320 lab photographs of pressed leaves for training and 2,075 field phone photographs for testing (2,072 usable). Folds are grouped by the 1,277 physical leaves that the file names identify, after removing 212 exact duplicates. Unlike DIMPSAR, the field photographs show one leaf on plain paper, with little clutter. Masks for both sets come from a label-free background-colour mask that keeps the most leaf-coloured component; an automatic colour check flags 5.5% of lab and 1.8% of field masks, and on inspection the real failures are concentrated on needle-leaved species (*Abies concolor*, *Cryptomeria japonica*) and lab photographs that include a colour chart. They were kept. Two details changed after registration, before any model was trained and without looking at results: the mask now keeps the most leaf-coloured rather than the largest component, after inspection showed that the largest was often the colour chart, and folds are grouped by the leaf identifiers of the file names alone, because hash-based re-shot links merged unrelated leaves on these mostly blank frames. We trained plain, patch pooling, mask pooling, strip pooling and SVA (ConvNeXt-T, same recipe, 15 epochs, 5 folds × 2 seeds) and report 30-way accuracy with the same analysis, i.e. one Holm family over raw and masked photographs (table 4). Accuracies are therefore 30-way here and 20-way on DIMPSAR, and absolute values are not comparable across the two datasets; the contrasts are.

All models reach 99.3–99.4% on lab photographs. On raw field photographs, plain reaches 45.9%, patch and mask pooling 47.4% and 47.6%, strip pooling 50.1% and SVA 49.5%. The predictions fare as follows. (1) *Total gain*, partly confirmed: strip pooling beats plain by +4.1 points (10/10 runs, Holm $p = 0.006$), SVA by +3.6 (9/10), which is not significant after correction. (2) *Leaf location is smaller than on DIMPSAR*, confirmed: mask minus patch pooling is +0.1 [−1.8, +2.0], against +4.4 on DIMPSAR at the time of registration (+4.0 with all 25 runs); with little background to

Table 3: Standard vs. strong (background compositing + blur + noise) augmentation, ConvNeXt-T, 5 folds × 3 seeds, %. MedLeaf-C: seed 0.

Augmentation	Model	MedLeaf-C	Field (mask)	Field (raw)
standard	Plain	81.1 ± 1.1	35.1 ± 1.3	33.5 ± 1.2
	Mask pooling	84.7 ± 0.6	41.0 ± 1.0	41.2 ± 1.9
	Strip pooling	87.6 ± 0.8	42.9 ± 1.6	47.8 ± 2.0
	SVA (ours)	88.6 ± 0.5	43.1 ± 2.8	49.3 ± 2.8
strong	Plain	96.5 ± 0.4	43.1 ± 1.3	42.0 ± 1.7
	Mask pooling	97.5 ± 0.4	49.3 ± 1.1	50.1 ± 1.5
	Strip pooling	97.9 ± 0.2	49.3 ± 1.3	53.4 ± 1.7
	SVA (ours)	97.9 ± 0.1	49.4 ± 1.7	55.2 ± 2.0

discount, knowing where the leaf is adds nothing. (3) *Localisation still helps*, not confirmed: strip minus mask pooling is +2.5 (9/10) but its interval includes zero (Holm $p = 0.63$); with ten runs the 95% interval is about ±3 points wide, so the replication cannot distinguish a small localisation effect from none. (4) *Measured geometry adds little*, confirmed: SVA minus strip pooling is −0.6, equivalent to zero within ±3 points ($p = 0.046$). In sum, two predictions were confirmed, one held for strip pooling but not for SVA, and one was not confirmed. The total architectural gain is a third of that on DIMPSAR (4 vs. 13 points), as expected if most of it comes from coping with field clutter. Whitening the field backgrounds lowers every model to 40.7–43.5%, probably because LeafSnap's field paper already resembles the lab paper; no masked-photo contrast is significant.

Extension to five seeds (not pre-registered). Because ten runs could not resolve a 3-point effect, we added seeds 2–4 after the pre-registered analysis was complete; the pre-registered verdicts above are those of seeds 0–1 and are unchanged. With 25 runs per model (table 11, same Holm family), plain reaches 44.9% on raw field photographs, mask pooling 47.3%, strip pooling 50.6% and SVA 48.9%. Both total gains are now significant (strip pooling +5.7 [+3.2, +8.2], 25/25, Holm $p < 0.001$; SVA +4.0, 24/25, Holm $p = 0.010$), leaf location remains null (+0.1, 13/25), and localisation is +3.3 [+0.3, +6.3] (23/25), whose interval excludes zero but which does not survive the Holm correction ($p = 0.21$). SVA minus strip pooling is −1.7 [−4.2, +0.8] (5/25 runs in favour of SVA); equivalence within ±3 points is no longer shown ($p = 0.15$), and if anything measured geometry costs a little accuracy here.

Two further checks were run after the replication and were not pre-registered (seed 0, 5 folds). Training for 40 instead of 15 epochs leaves both models where they were (plain 45.9%, strip pooling 50.7%, +4.9 points, 5/5 folds), so the short schedule neither creates nor hides the gain. BioCLIP 2 reaches 67.9% on raw field photographs (61.9% masked), about 18 points above the best ConvNeXt-T model, and strip pooling adds nothing to it (+0.0, 3/5 folds), as on DIMPSAR.

6.5 The gains depend on the backbone

On EfficientNet-B0 (3 seeds) all four structural models gain only 1.5–3 points on raw photographs (plain 26.8%, SVA 29.8%, +3.0 [−2.6, +8.6], 11/15) and nothing on masked ones. On ViT-S, localised pooling may still help (strip 62.7%, slab-only 64.8% vs. plain 60.4%, one seed), but SVA does not (60.7%). BioCLIP 2 reaches 81.4% on raw photographs without any module, about 30 points above the best ImageNet ConvNeXt-T model, and all variants lie within 0.5 points of each other. A biology foundation model thus closes most of the studio-to-field gap on its own. The

Table 4: Pre-registered contrasts on LeafSnap30 field photographs (ConvNeXt-T, 10 matched runs, 30-way accuracy). Columns as in table 2; one Holm family.

Set	Photos	Contrast	Δ (runs won)	95% CI, runs	95% CI, runs+species	Holm p
pre-registered	raw	strip − plain	+4.1 (10/10)	[+2.3, +5.9]	[+0.3, +8.8]	0.006
		SVA − plain	+3.6 (9/10)	[+0.3, +6.9]	[-0.1, +7.9]	0.34
		mask pooling − patch pooling	+0.1 (6/10)	[-1.8, +2.0]	[-2.6, +2.2]	1.00
		strip − mask pooling	+2.5 (9/10)	[-0.5, +5.5]	[-0.2, +5.4]	0.63
		SVA − strip	-0.6† (3/10)	[-3.5, +2.4]	[-3.5, +2.3]	1.00
	masked	strip − plain	+1.7 (8/10)	[-1.9, +5.2]	[-1.7, +5.8]	1.00
		SVA − plain	+2.8 (9/10)	[+0.2, +5.5]	[+0.1, +6.3]	0.34
		mask pooling − patch pooling	+1.2 (8/10)	[-0.4, +2.9]	[-0.7, +3.5]	0.77
		strip − mask pooling	-1.1 (3/10)	[-4.3, +2.1]	[-3.8, +1.2]	1.00
		SVA − strip	+1.2 (7/10)	[-1.3, +3.7]	[-1.7, +4.3]	1.00

advantage is specific to biological pre-training rather than to large-scale pre-training as such: a DINOv2 ViT-S/14, self-supervised on 142M general images, reaches only 52.5% (one seed), and on it mask pooling hurts (48.0%) while strip pooling is neutral (53.0%). For both ViTs, masked photographs are 8–22 points *harder* than raw ones, whereas for the CNNs the two differ by at most about five points; the whitened background and the masker's errors apparently remove context these models use, so we treat raw photographs as the primary field test.

6.6 Corruptions and cue conflict

On MedLeaf-C (fig. 4) the structural models are more robust on ConvNeXt-T (seed 0: plain 81.1%, patch pooling 84.3%, mask pooling 84.7%, strip 87.6%, SVA 88.6%, slab-only 89.4%), with the gains coming from noise, blur, JPEG and low-resolution corruptions; colour, occlusion and background corruptions are near ceiling for all models, and on the other backbones all variants are within about one point. This comparison favours the mask-based models, however: under corruption they receive the *true* leaf mask, while the plain and patch-pooling models receive none. Its mask-free part, patch pooling over plain (+3.2 points on seed 0), is about as large as the further step from mask pooling to localised pooling, so we read MedLeaf-C as consistent with, but not evidence for, a texture-robustness mechanism. Shape bias remains low everywhere (7–14%): when a silhouette is filled with another species' lamina, all models mostly follow the lamina. Localised pooling raises ConvNeXt-T shape bias only slightly (plain 6.5%, strip 9.1%, SVA 9.0%; seed 0), and BioCLIP 2 is the most shape-biased backbone (14%). The localised models are therefore not shape-based.

6.7 Inside SVA: ablations

Table 10 varies one ingredient of SVA at a time (ConvNeXt-T). A single-seed screen on folds 0–2 (lower block) was followed by the four most informative ablations on all 5 folds × 3 seeds (upper block), compared with SVA on the same runs. The screen overstated every effect, because its SVA reference happened to be strong (53.1% raw vs. 49.3% over all seeds). With three seeds, a coarser sweep costs 5.1 [1.1, 9.1] raw points (15/15) and unit cell weights 4.5 [−0.5, 9.5] (13/15); Euler tokens transfer slightly worse than area and perimeter (−2.6, 12/15, n.s.), although they are the only ConvNeXt-T tokens that make the model respond more to holes in the blade than to margin edits (section F); and restricting attention to the slab is not needed (global attention 49.3% vs. 49.3%). These effects concern the internal design of SVA; as shown above, the full module is not better than localised pooling without any measurement. The vein head reproduces its ridge-filter

supervision closely on held-out leaves (Pearson $r = 0.96$, Dice 0.74–0.75 inside the blade), which measures agreement with the pseudo-label, not with traced veins.

6.8 Leakage audit

Under random image-level splits that keep the duplicates, 5–11 test images per fold (1.4–3.0%) have an exact duplicate or a re-shot of the same leaf in the training part; under our grouped protocol there are none. On this dataset the leak does not measurably inflate accuracy: a DeepHerb-style reproduction (Xception features with an RBF-SVM [2]) reaches $98.5 \pm 0.9\%$ with random splits and $98.4 \pm 0.7\%$ with grouped splits, the dual-branch RGB + vein-skeleton baseline [8] reaches 99.95% under both, and our plain and SVA models reach 100% under random splits as under grouped ones. We therefore present the grouped protocol as hygiene that removes a known leak rather than as evidence that earlier results were inflated; the saturation itself is the more important observation.

7 Discussion

What closes the studio-to-field gap. For an ImageNet ConvNeXt-T trained on studio leaves, the architectural gain on raw field photographs has three identifiable parts. An extra branch on mid-level features helps a little (3.5 points, not significant on raw photographs, 4.4 on whitened ones). Pooling that branch over the leaf adds about 4 points, and only when the field background is present: on photographs whose background has been whitened it adds almost nothing, so this part is the model discounting the background. Pooling locally within the leaf adds another 5–6 points on raw photographs but little on whitened ones, and it barely changes shape bias. Why localisation helps when clutter is present remains open; a plausible reading is that pooling small regions separately keeps background that leaks through an imperfect mask from contaminating the whole leaf representation, but we have not tested this. Mask pooling and localisation both survive strong background-compositing augmentation, which lifts every model but replaces neither.

What does not. We expected measured leaf geometry to matter: an object-centred sweep, per-slab area and boundary length of the blade and venation, and attention restricted to the slab being measured. Fusing the measurements globally is equivalent to the plain model within ± 3 points (two one-sided tests, $p < 0.01$ on both photo types). Attached to localised tokens, they improve on strip pooling and slab-only attention by $+0.5$ and $+1.7$ points on raw photographs, with intervals that exclude large benefits but not benefits of up to 3–5 points; object-centred slabs and image-axis strips perform the same, and restricting attention to the measured slab can be replaced by global attention. Within SVA, a fine sweep and learned weights matter, but only relative to weaker versions of SVA itself, and Euler tokens make a model sensitive to holes in the blade yet transfer worse. Within the resolution of our design, the effects we can detect concern *where* features are pooled, not measurements of the leaf. We report SVA because a well-controlled negative result for a natural and increasingly popular idea, fusing topological and geometric measurements with deep features, is useful.

Practical recommendations. (i) Evaluate leaf classifiers on field photographs; studio accuracy is saturated and uninformative (98–100% for every model). (ii) With an ImageNet CNN, pool over the leaf and locally within it, e.g. with strip pooling, which is simpler and cheaper than SVA, and combine it with background-compositing augmentation; together they raised field accuracy from 34% to 53–55%. Note that a leaf mask is needed at test time, for cropping and for these models;

our masker costs about as much as the classifier (section E). (iii) Where a biology foundation model such as BioCLIP 2 can be used, use it: it reaches 81% without any module, and none of the modules changes it. Recommendations (i) and (ii) matter where it cannot, e.g. on-device or offline inference: its ViT-L/14 visual tower has about ten times the parameters of ConvNeXt-T.

Does it generalise? The LeafSnap30 replication, with predictions fixed in advance, supports the clutter account. Where field leaves lie on plain paper, the leaf-location effect disappears, the total gain shrinks from 13 to 4 points, and measured geometry again adds nothing; localised pooling stays ahead of mask pooling (+2.5) but not significantly. With five seeds (beyond the pre-registration) the total gain is +5.7 for strip pooling and +4.0 for SVA, both significant, and localisation is +3.3 in 23 of 25 runs yet still not significant after correction. Localisation is thus the less certain of the two ingredients outside DIMPSAR. BioCLIP 2 again outperforms every ImageNet model (67.9%) and again gains nothing from strip pooling. The replication also shows a limit of the mask-based recipe: label-free masks fail most often on needle-leaved conifers, whose leaves are thin and small in the frame, so leaf location and localised pooling presuppose broad leaves that can be segmented.

Limitations. (i) Two studio-to-field pairs, and only DIMPSAR has cluttered field backgrounds; the medicinal pair has 20 shared species. Field labels come from vernacular folder names that we mapped to species and checked visually but not with a botanist; residual label noise would lower all models alike. The 20-way accuracy uses the knowledge of which species can occur; 30-way accuracy is lower but keeps the ordering of the models. The only high-hazard species of the studio set, *Nerium oleander*, is not among the field species, so we could not evaluate confusions with a toxic look-alike in the field. (ii) The field test set depends on our masker for every model: it centres the crops and excluded 5.6% of the photographs, probably among the hardest, and the mask-based models also use its masks inside the network. Its accuracy on field photographs could not be measured without ground-truth masks. (iii) ConvNeXt-T results use five seeds (LeafSnap30 two pre-registered, five in total), EfficientNet-B0 and the key ablations three, the ViTs one; differences of 2–3 points between localised models are not resolved, and the ViT results are indicative only. (iv) One fixed training recipe was used for all models of a backbone; a recipe tuned per model could shift individual numbers. (v) The mask- and patch-pooling controls were added after a first analysis; on DIMPSAR their contrasts confirm a suspected confound rather than test a pre-registered hypothesis (the LeafSnap30 replication was pre-registered). (vi) The venation field is learned from ridge-filter pseudo-labels and is not validated against traced veins.

8 Conclusion

Medicinal-leaf classifiers that are perfect on studio photographs transfer poorly to field photographs. In a controlled study with matched models, multiple seeds and corrected tests, we find that for an ImageNet CNN the architectural part of the remedy is simple: pool over the leaf, which lets the model discount the field background, and pool locally within it. Each step is worth about 4–6 points on raw field photographs, and both add to background-compositing augmentation. A pre-registered replication on LeafSnap30, where field leaves lie on plain paper, finds a smaller gain and no leaf-location effect, as predicted, and again no benefit of measured geometry. Measured leaf geometry, introduced here as sweep-valuation attention with appearance-weighted area, boundary and Euler measurements along an object-centred sweep, adds little if anything beyond localised pooling, and a biology foundation model makes all such modules unnecessary. We release the

leakage-audited protocol, the field mapping, the corruption, cue-conflict and topology probes, and all code and run outputs.

Code and data availability

All code, configurations, unit tests, the outputs of every training run (metrics, probe results, per-image predictions and logs), the statistical analyses and the LaTeX sources of this paper are publicly available at `https://github.com/Devguru-codes/research_paper_leaf`. Every number reported here is regenerated from those run outputs by the scripts in the repository, and the properties stated in section 4 are verified by its unit tests (section B). The image datasets are not redistributed; they are available from their original sources (section 3, section 6.4) [26, 27, 33].

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A Details of sweep-valuation attention

Filtration increments. For each of K=32 directions θ, a cell c has height $h_\theta(c) = \max_{v\in c}\langle x_v, \theta\rangle$, and with the sigmoid relaxation of DECT [13] the soft membership of height h in slab k of T=16 is

$$s_k(h) = \sigma\big(\lambda(t_k - h)\big) - \sigma\big(\lambda(t_{k-1} - h)\big), \qquad \textstyle\sum_k s_k(h) = 1, \tag{2}$$

with uniform thresholds and $t_0 = -\infty$, $t_T = +\infty$. The weighted cell counts entering slab k, $\Delta H_d(\theta, k) = \sum_c w_d(c)\, s_k(h_\theta(c))$, give the area and boundary-length increments $\Delta A = \Delta H_2$ and $\Delta P = 2\Delta H_1 - 4\Delta H_2$ (and $\Delta\chi$ for the Euler ablation), which telescope to the totals of the leaf. The token feature $\phi(\theta, k)$ collects these increments for both fields and their cumulative sums up to slab k (8 values, signed-log scaled): the local width and boundary complexity of the slab, and its position in the leaf.

Slab-restricted attention. Token (θ, k) has query $q_{\theta k} = \mathrm{MLP}(\phi) + e_k$ with a learned slab embedding e_k and attends to the patch tokens z_p only within the *same* slab:

$$\mathrm{attn}_{(\theta,k)}(p) \propto \exp\Big(\tfrac{\langle W_Q q_{\theta k}, W_K z_p\rangle}{\sqrt{d}} + \log\big(s_k(\langle x_p, \theta\rangle)\, \bar{m}(p) + \epsilon\big)\Big), \tag{3}$$

where $\bar{m}(p)$ is the leaf fraction of patch p. The 512 tokens are mixed by a two-layer transformer with a cyclic relative position bias over directions and mean-pooled.

Valuations. For a binary set, $A = H_2$ counts unit squares, $P = 2H_1 - 4H_2$ is the boundary length in pixel units (every square contributes four edges, and an edge shared by two squares is interior) and $\chi = H_0 - H_1 + H_2$ is the Euler characteristic. In the continuum these are the planar intrinsic volumes, which by Hadwiger's theorem [22, 23] span the motion-invariant valuations on *convex* bodies; we use this only to motivate the basis. Leaves are not convex, the digital perimeter is the anisotropic ℓ_1 boundary length (it overestimates a straight boundary at angle α to the grid by $|\cos\alpha| + |\sin\alpha|$, up to $\sqrt{2}$ [20]), and on the fuzzy complex (1) the quantities are weighted counts rather than valuations of a set.

Sweep and token. Pixel centres are expressed in a frame centred at the blade centroid and scaled by its root-mean-square radius, which makes all quantities invariant to translation and isotropic scale. Thresholds $t_1, \dots, t_{T-1}$ are uniform on $(-R, R)$ with R=2, and $\lambda = \kappa T/2R$ with κ=4. The token feature is $\phi(\theta, k) = \mathrm{slog}[\Delta A_\mathrm{b}, \Delta P_\mathrm{b}, \Delta A_\mathrm{v}, \Delta P_\mathrm{v}, \sum_{j \le k}(\cdot)]$ with $\mathrm{slog}(x) = \mathrm{sign}(x)\log(1 + |x|)$. The Euler variant replaces it by $[\Delta\chi_\mathrm{b}, \Delta\chi_\mathrm{v}, \chi_\mathrm{b}(t_k), \chi_\mathrm{v}(t_k)]$ and the Hadwiger variant uses all three valuations (12 features). The last layer of each weight MLP g is initialised so that $a \approx 0.88\,m$, and the vein head also receives task gradients through SVA. In contrast to fixed intensity weights (WECT [14]), the weights are learned.

Efficient computation. Directions are fixed and coordinates come from the mask, so the heights carry no gradient; only the weights do. We hard-bin heights into 256 fine bins, scatter-add cell weights per dimension into a per-direction histogram and apply the slab kernel (2) evaluated at bin centres with one matrix product (built in double precision). This needs no sorting and is exact in the telescoping and sharp limits (section B).

Attention and training. In (3), $\epsilon = 10^{-4}$. The attention output is added to the query with a mass gate $M/(M + 0.5)$, $M = \sum_p s_k \bar{m}$, so that slabs outside the leaf contribute little. The sweep transformer's learned relative position bias depends only on $(\Delta\theta \bmod K, \Delta k)$. Training minimises cross-entropy (label smoothing 0.1) plus 0.2× the masked BCE of the vein head, which all models share.

Rotation. For unit weights, rotating the input by a multiple of 90° cyclically shifts all increments by $K/4$ directions (section B); the appearance-weighted signal and the full model inherit the rotation behaviour of the backbone and are evaluated empirically. Rotation-equivariant ECT tokenisation as such is not our contribution [7].

B Unit tests of the structural signals

All properties used in section 4 are checked by automated tests that ship with the code (35 tests, all passing; the list covers the properties used in this paper).

- **Exact Euler characteristic.** On toy shapes (disk, annulus, two disks, a lens-shaped leaf, and the lens with two holes; $\chi = 1, 0, 2, 1, -1$) the cubical $\chi = V - E + F$ of the binary set equals the value computed independently with scikit-image (4-connected foreground).
- **Telescoping.** For every direction, $\sum_k \Delta\chi(\theta, k)$, $\sum_k \Delta A(\theta, k)$ and $\sum_k \Delta P(\theta, k)$ equal the total valuation of the shape, to numerical precision.
- **Partition of unity.** Slab memberships (2) sum to one for every height.

- **Sharp limit.** With binary weights and large κ, the relaxed increments equal those of a hard sweep that assigns each cell to the slab containing its height.
- **Closed form.** For an $n \times n$ block of pixels, the totals are $(A, P, \chi) = ((n-1)^2, 4(n-1), 1)$.
- **Consistency.** The Euler column of the Hadwiger increments equals the separately implemented Euler increments.
- **Rotation.** For unit weights, rotating the input by 90° cyclically shifts $\Delta\chi$, ΔA and ΔP by $K/4$ directions. This follows from $\langle \mathcal{R}x, \theta \rangle = \langle x, \mathcal{R}^{-1}\theta \rangle$, the equivariance of the centroid/RMS frame and the fact that a 90° grid rotation maps cells of each dimension bijectively onto cells of the same dimension; for rotations by $2\pi/K$ it holds up to grid resampling.
- **Mask pooling.** The mask-pooling control is unchanged when patch tokens outside the leaf mask are altered.
- **Fuzzy subcomplex.** Learned cell weights satisfy $w(\text{coface}) \leq w(\text{face})$, and gradients reach the per-pixel weights.
- **Splits.** Under the grouped protocol no exact duplicate or re-shot group crosses a fold boundary and every image is tested exactly once; under the random protocol leaks occur.

C Species and taxonomy

D Field-photo test set

E Computational cost

F Topology sensitivity

The topology-destruction probe compares edits that change the blade's Euler characteristic (three interior holes, $\Delta\chi = -3$) with edits that remove the same area from the margin ($\Delta\chi = 0$). The two edit types are not perceptually identical (median LPIPS 0.045 vs. 0.030), so we compare them only within LPIPS strata that contain both (10 quantile bins, at least five edits of each kind) and report the stratum-weighted log-ratio of the prediction change (KL divergence), $\log(\mathrm{KL}_{\text{holes}}/\mathrm{KL}_{\text{bites}})$, with a bootstrap 95% interval (table 9). Every ImageNet ConvNeXt-T and ViT-S model responds *less* to holes than to margin bites, even though the holes are the more visible edit. The only ConvNeXt-T model that responds more to holes is the Euler-token variant of SVA (+0.73 [+0.54, +0.96]), so measured topology does change what a model is sensitive to. BioCLIP 2 responds more to holes for every variant (+1.3 to +1.6), i.e. the foundation model is topology-sensitive by itself. Topological sensitivity can thus be induced, but it did not help species transfer (section 6.7). Edits of the venation could not be matched perceptually (only 25% of vein edits fall inside the LPIPS range of their controls), so we draw no conclusion from them.

G Additional results

Table 5: The 30 species (APG IV order and family) and the number of images after de-duplication.

Order / family	Species	Name	n
Brassicales / Brassicaceae	*Brassica juncea*	Indian Mustard	34
Brassicales / Moringaceae	*Moringa oleifera*	Drumstick	77
Caryophyllales / Amaranthaceae	*Amaranthus viridis*	Arive-Dantu	122
Caryophyllales / Basellaceae	*Basella alba*	Basale	103
Fabales / Fabaceae	*Pongamia pinnata*	Indian Beech	61
Fabales / Fabaceae	*Trigonella foenum-graecum*	Fenugreek	36
Gentianales / Apocynaceae	*Carissa carandas*	Karanda	64
Gentianales / Apocynaceae	*Nerium oleander*	Oleander	62
Gentianales / Apocynaceae	*Tabernaemontana divaricata*	Crape Jasmine	56
Lamiales / Lamiaceae	*Mentha sp.*	Mint	97
Lamiales / Lamiaceae	*Ocimum tenuiflorum*	Tulsi	52
Lamiales / Lamiaceae	*Plectranthus amboinicus*	Mexican Mint	48
Lamiales / Oleaceae	*Jasminum sp.*	Jasmine	71
Lamiales / Oleaceae	*Nyctanthes arbor-tristis*	Parijata	40
Malvales / Malvaceae	*Hibiscus rosa-sinensis*		43
Malvales / Muntingiaceae	*Muntingia calabura*	Jamaica Cherry-Gasagase	56
Myrtales / Lythraceae	*Punica granatum*	Pomegranate	79
Myrtales / Myrtaceae	*Psidium guajava*	Guava	65
Myrtales / Myrtaceae	*Syzygium cumini*	Jamun	39
Myrtales / Myrtaceae	*Syzygium jambos*	Rose Apple	56
Piperales / Piperaceae	*Piper betle*	Betel	48
Rosales / Moraceae	*Artocarpus heterophyllus*	Jackfruit	56
Rosales / Moraceae	*Ficus auriculata*	Roxburgh fig	50
Rosales / Moraceae	*Ficus religiosa*	Peepal Tree	63
Santalales / Santalaceae	*Santalum album*	Sandalwood	58
Sapindales / Anacardiaceae	*Mangifera indica*	Mango	62
Sapindales / Meliaceae	*Azadirachta indica*	Neem	60
Sapindales / Rutaceae	*Citrus limon*	Lemon	57
Sapindales / Rutaceae	*Murraya koenigii*	Curry	60
Zingiberales / Zingiberaceae	*Alpinia galanga*	Rasna	50

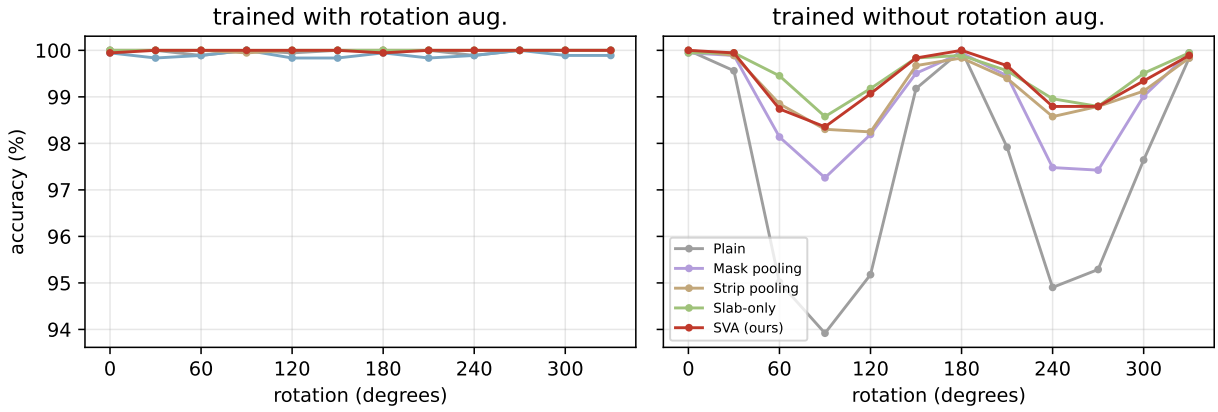


Figure 3: Accuracy versus test-time rotation (ConvNeXt-T, seed 0, mean over 5 folds), trained with (left) and without (right) rotation augmentation.

Table 6: DIMPSAR folders used for the field test, the species they were mapped to (checked visually), and the number of images in which our masker found a leaf. Kannada *Arali* denotes *Ficus religiosa*, not *Nerium oleander*.

DIMPSAR folder	Species	Family	Images used
Jackfruit	*Artocarpus heterophyllus*	Moraceae	110
Neem	*Azadirachta indica*	Meliaceae	113
Malabar Spinach	*Basella alba*	Basellaceae	79
Lemon	*Citrus limon*	Rutaceae	120
Arali	*Ficus religiosa*	Moraceae	89
Hibiscus	*Hibiscus rosa-sinensis*	Malvaceae	113
Jasmine	*Jasminum sp.*	Oleaceae	48
Mango	*Mangifera indica*	Anacardiaceae	103
Mint	*Mentha sp.*	Lamiaceae	122
Drumstick	*Moringa oleifera*	Moringaceae	26
Gasagase	*Muntingia calabura*	Muntingiaceae	77
Curry	*Murraya koenigii*	Rutaceae	159
Parijatha	*Nyctanthes arbor-tristis*	Oleaceae	61
Tulsi	*Ocimum tenuiflorum*	Lamiaceae	171
Betel	*Piper betle*	Piperaceae	112
Doddpathre	*Plectranthus amboinicus*	Lamiaceae	131
Honge	*Pongamia pinnata*	Fabaceae	110
Guava	*Psidium guajava*	Myrtaceae	121
Pomoegranate	*Punica granatum*	Lythraceae	73
Nerale	*Syzygium cumini*	Myrtaceae	62

Table 7: Field photographs per species, sorted by the accuracy of the plain model: DIMPSAR photographs, photographs excluded because our masker found no plausible leaf, median share of the crop covered by the leaf, and 20-way accuracy (%) on raw photographs (ConvNeXt-T, mean over all runs). The plain model fails mostly on species with ovate to elliptic leaves without a distinctive outline; the exclusions concentrate on the compound-leaved *Moringa* and *Azadirachta*. The localisation gain is unrelated to leaf share (Spearman $\rho = -0.06$, $p = 0.81$).

Species	Photos	Excluded	Leaf (%)	Plain	Strip	SVA
Jasminum	49	1	18	1.7	5.5	5.3
Nyctanthes arbor-tristis	66	5	14	4.5	7.7	4.2
Pongamia pinnata	113	3	12	12.1	24.0	16.8
Mangifera indica	103	0	18	12.9	21.3	19.7
Citrus limon	123	3	10	13.2	25.2	42.6
Syzygium cumini	62	0	19	14.6	7.9	2.1
Azadirachta indica	132	19	3	16.3	45.4	53.3
Mentha	135	13	13	21.8	54.0	45.1
Artocarpus heterophyllus	110	0	22	22.1	34.5	31.3
Punica granatum	75	2	9	24.5	67.1	63.9
Muntingia calabura	79	2	18	28.1	53.0	64.9
Hibiscus rosa-sinensis	118	5	6	30.3	40.3	49.8
Moringa oleifera	56	30	4	35.1	33.5	35.5
Plectranthus amboinicus	142	11	15	37.0	65.7	61.0
Psidium guajava	128	7	9	41.2	52.4	54.9
Ocimum tenuiflorum	177	6	5	43.6	50.1	49.5
Basella alba	79	0	20	54.4	63.5	61.6
Murraya koenigii	168	9	8	57.4	54.1	49.3
Piper betle	114	2	34	77.2	84.1	86.5
Ficus religiosa	89	0	26	80.1	91.4	95.1

Table 8: Parameters, GFLOPs (one image at training resolution) and median batch-1 latency on a laptop RTX 3050 GPU and on CPU. Mask pooling and strip pooling add almost no cost; the sweep-based models roughly double CPU and GPU latency, mostly because of the histogram operations.

Backbone	Model	Params (M)	Added (M)	GFLOPs	GPU ms	CPU ms
ConvNeXt-T	Plain	27.9	0.12	27.9	16	341
	Mask pooling	28.1	0.23	28.0	21	347
	Strip pooling	28.9	1.11	28.0	18	318
	Slab-only	29.1	1.26	29.3	26	661
	SVA	29.1	1.30	29.3	30	621
	Global fusion	29.6	1.79	28.1	28	411
EfficientNet-B0	Plain	4.1	0.09	3.2	23	270
	Strip pooling	5.0	1.03	3.2	25	294
	Slab-only	5.1	1.12	4.3	31	386
	SVA	5.2	1.16	4.3	49	431
	Global fusion	5.8	1.75	3.3	39	348
ViT-S/16	Plain	22.1	0.27	29.6	16	462
	Strip pooling	23.1	1.27	29.7	17	510
	Slab-only	23.2	1.42	31.2	22	479
	SVA	23.3	1.46	31.3	30	635
	Global fusion	23.8	1.96	30.1	32	569
Leaf masker (448^2)	–	28.0	–	38.2	27	307

Table 9: LPIPS-matched response to topology-changing holes relative to area-matched margin bites, $\log(\mathrm{KL}_{\mathrm{holes}}/\mathrm{KL}_{\mathrm{bites}})$ with bootstrap 95% CI (seed 0, 5 folds). Negative: the model reacts more to margin geometry than to topology.

Backbone	Model	log-ratio [95% CI]
ConvNeXt-T	Plain	-1.22 [-1.47, -1.02]
	Global fusion	-0.95 [-1.15, -0.75]
	Strip pooling	-0.90 [-1.02, -0.77]
	Slab-only	-0.96 [-1.11, -0.81]
	SVA	-1.63 [-1.76, -1.48]
	Euler token (ablation)	+0.73 [+0.54, +0.96]
ViT-S/16	Plain	-0.35 [-0.51, -0.22]
	Global fusion	-0.62 [-0.76, -0.49]
	Strip pooling	-0.84 [-1.03, -0.65]
	Slab-only	-0.70 [-0.88, -0.52]
	SVA	-0.42 [-0.60, -0.25]
BioCLIP 2	Plain	+1.63 [+1.50, +1.76]
	Global fusion	+1.47 [+1.34, +1.60]
	Strip pooling	+1.34 [+1.23, +1.45]
	Slab-only	+1.36 [+1.23, +1.50]
	SVA	+1.60 [+1.46, +1.74]

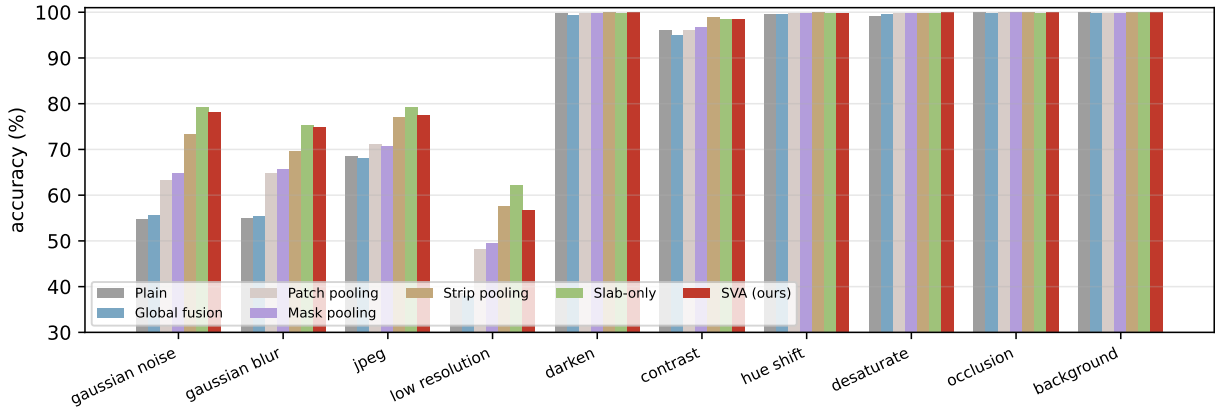


Figure 4: MedLeaf-C accuracy per corruption family (ConvNeXt-T, seed 0, severity-averaged). Mask-based models receive the true leaf mask under corruption.

Table 10: Ablations of SVA (ConvNeXt-T, %), each block sorted by raw field accuracy and compared with SVA on the same runs. Rotation is evaluated on seed 0 only.

Variant	Clean	Rot. min	Field (mask)	Field (raw)
5 folds × seeds 0–2				
SVA (full)	100.0 ± 0.1	99.9 ± 0.1	43.1 ± 2.8	49.3 ± 2.8
attention over all leaf patches	100.0 ± 0.0	100.0 ± 0.0	44.0 ± 2.6	49.3 ± 2.4
Euler token	100.0 ± 0.0	100.0 ± 0.0	42.6 ± 1.4	46.8 ± 2.1
unit cell weights	100.0 ± 0.1	99.8 ± 0.2	41.3 ± 1.4	44.8 ± 1.6
coarse sweep (K=16, T=8)	100.0 ± 0.1	99.9 ± 0.2	41.5 ± 2.1	44.2 ± 3.2
seed 0, folds 0–2				
SVA (full)	99.9 ± 0.2	99.9 ± 0.2	47.0 ± 0.7	53.1 ± 2.7
softer relaxation (κ=1)	100.0 ± 0.0	99.9 ± 0.2	46.8 ± 1.1	52.5 ± 2.1
attention over all leaf patches	100.0 ± 0.0	100.0 ± 0.0	47.1 ± 1.1	52.0 ± 2.5
weights permuted within the leaf	100.0 ± 0.0	100.0 ± 0.0	44.5 ± 0.8	50.8 ± 1.1
venation field only	99.9 ± 0.2	99.8 ± 0.2	45.3 ± 0.6	50.7 ± 2.5
sharper relaxation (κ=16)	100.0 ± 0.0	99.9 ± 0.2	46.5 ± 2.6	50.6 ± 3.8
blade only	100.0 ± 0.0	99.9 ± 0.2	43.8 ± 2.0	48.4 ± 4.1
coarse sweep (K=16, T=8)	100.0 ± 0.0	99.9 ± 0.2	43.2 ± 1.2	46.8 ± 1.0
Euler token	100.0 ± 0.0	100.0 ± 0.0	43.0 ± 0.7	45.7 ± 2.8
Hadwiger token (area, perimeter, Euler)	100.0 ± 0.0	100.0 ± 0.0	38.9 ± 1.1	44.3 ± 1.5
unit cell weights	99.9 ± 0.2	99.8 ± 0.2	41.5 ± 1.2	43.5 ± 0.7

Table 11: LeafSnap30 contrasts extended to five seeds (ConvNeXt-T, 25 matched runs, 30-way accuracy; not pre-registered, see section 6.4). Columns as in table 2; one Holm family.

Set	Photos	Contrast	Δ (runs won)	95% CI, runs	95% CI, runs+species	Holm p
extension	raw	strip − plain	+5.7 (25/25)	[+3.2, +8.2]	[+2.5, +9.7]	0.001
		SVA − plain	+4.0 (24/25)	[+1.7, +6.2]	[+0.7, +8.0]	0.010
		mask pooling − patch pooling	+0.1 (13/25)	[-1.8, +1.9]	[-2.1, +1.8]	1.00
		strip − mask pooling	+3.3 (23/25)	[+0.3, +6.3]	[+1.4, +5.7]	0.21
		SVA − strip	-1.7 (5/25)	[-4.2, +0.8]	[-3.7, +0.0]	0.90
	masked	strip − plain	+3.6 (23/25)	[+0.2, +7.1]	[+0.9, +7.1]	0.25
		SVA − plain	+3.3 (23/25)	[+0.7, +5.9]	[+1.2, +5.9]	0.13
		mask pooling − patch pooling	+0.8 (18/25)	[-0.6, +2.2]	[-0.7, +2.6]	0.92
		strip − mask pooling	+0.7 (17/25)	[-2.0, +3.4]	[-1.1, +2.4]	1.00
		SVA − strip	-0.4† (11/25)	[-3.2, +2.5]	[-2.8, +1.9]	1.00